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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025

Time: 03 Hours

Code: 100202 ***Subject: Engineering Mathematics - II*** ***Full Marks: 70***

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*

Q.1 Choose the correct option/answer the following (Any seven question only): **[14]**

(a) If $f(z)=u(x,y)+iv(x,y)$ is analytic function, then $f'(z)=$

(i) $u/x- iv/x$ (iii) $v/y- iv/x$
(ii) $u/x- iv/y$ (iv) $u/x- iv/y$

(b) For an analytic function $f(z)=u(x,y)+i\ v(x,y)$, if the real part $u(x,y)=x^2-y^2$ is given, what is the imaginary part $v(x,y)$?

(i) $2xy+c$ (iii) $x^2+y^2=c$
(ii) $xy+c$ (iv) $-2xy+c$

(c) The value of the integral $\oint_C|z|dz$, where C is the left half of the unit circle $|z|=1$ from $z=-i$ to $z=i$

(i) $2\pi i$ (iii) 0
(ii) πi (iv) $2i$

(d) The residue of $e^z/\cos\pi z$ at its pole $z=-1/2$ is

(i) $-e^{1/2}/\pi$ (iii) 0
(ii) $e^{1/2}/\pi$ (iv) $-e^{1/2}$

(e) The particular integral y_p of the differential equation $(D^2+4)y=\cos2x$, $D\equiv d/dx$ is

(i) $1/2\sin2x$ (iii) $1/4x\sin2x$
(ii) $1/2x\sin2x$ (iv) $1/2x\cos2x$

(f) By the method of undetermined coefficients, the particular integral y_p of the differential equation $(D^4+2D^3+D^2)\ y=12x^2$, $D\equiv d/dy$ is

(i) $a+bx+cx^2$ (iii) $x(a+bx+cx^2)$
(ii) $ax+bx^2+cx^3$ (iv) $x^2(a+bx+cx^2)$

(g) Let $\lim_{n\to\infty}(u_n)^{1/n}=l$, then the series $\sum_{n=1}^{\infty}u_n$ converges if

(i) $l > 1$ (iii) $l = 1$
(ii) $l < 1$ (iv) $l\geq1$

(h) $L(\sin at)=[a] / [s^2+a^2]$ when

(i) $s > a$ (iii) $s < 0$
(ii) $s < a$ (iv) $s > 0$

(i) If $L^{-1}([3s+8] / [s^2+4s+25])=e^{kt}[3\cos\sqrt{(21)}t+[2] / [\sqrt{(21)}]\sin\sqrt{(21)}t]$ then the value of k is

(i) -1 (iii) 2
(ii) 1 (iv) -2

(j) If $f(x)=\sqrt{([1-\cos x] / [2])}$, $0 < x < 2\pi$ then the value of a_0 is

(i) $4/\pi$ (iii) $\pi/4$
(ii) $2/\pi$ (iv) $\pi/2$

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Q.2 Solve all questions : **[14]**

(a) Find $\lim_{z \to 0}[[1] / [1-e^{1/x}]+ iy^2]$ if it exists.

(b) Check the continuity of the following function at $z=0$:
$f(z)=[\mathrm{Re}(z^2)] / [|z|^2]$, & zne0
0, & $z=0$

(c) Examine whether the Cauchy-Riemann equations are satisfied at the origin for the function given below or not:
$f(z)=[x^3(1+i)-y^3(1-i)] / [x^2+y^2]$, & zne0
0, & $z=0$
Also check, whether $f(z)$ is analytic at $z=0$ or not.

Q.3 Solve both questions : **[14]**

(a) If $f(\xi)=\int_C[3z^2+7z+1] / [z-\xi]dz$, where C is an ellipse $9x^2+16y^2=144$. Find the values of $f(3)$, $f'(1-i)$ and $f''(1-i)$.

(b) Evaluate the integral $\int_0^\infty[\sin ax] / [x(x^2+b^2)]dx$, $a>0$, $b>0$ by the use of Cauchy residue theorem.

Q.4 Solve both questions : **[14]**

(a) Find the Laurent's expansion for $f(z)=[7z-2] / [z^3-z^2-2z]$ in the region $1 <|z+1|<3$.

(b) Find the complete solution of $d^2y/dx^2-2dy/dx+2y=x+e^x\cos x$.

Q.5 Solve both questions : **[14]**

(a) Find the solution of $x^2d^2y/dx^2+xdy/dx+y=\log x\cdot\sin(\log x)$.

(b) Find the solution of $d^2y/dx^2-6dy/dx+9y=e^{3x}/x^2$ by the method of variation of parameters.

Q.6 Solve both questions : **[14]**

(a) Test the convergence of the series $\Sigma(1+1/n)^{-n^2}$.

(b) Test the convergence of the series $a/b+[a+x] / [b+x]+[a+2x] / [b+2x]+[a+3x] / [b+3x]+.....$

Q.7 Solve both questions : **[14]**

(a) Find Laplace transform of $t\cos at$.

(b) Evaluate the integral $\int_0^\infty te^{-3t}\sin t\ dt$, by the use of Laplace transform.

Q.8 Solve both questions : **[14]**

(a) Find the inverse Laplace transform of $\tan^{-1}(2/s^2)$.

(b) Find the solution of the following initial value problem by the use of Laplace transform:
$d^2y/dt^2+tdy/dt-y=0$, if $y(0)=0$ and $(dy/dt)_{t=0}=1$.

Q.9 Solve both questions : **[14]**

(a) Find the Fourier series expansion of the following function:
$f(x)= 0$, & $-\pi\le x\le 0$
$\sin x$, & $0\le x <\pi$
Hence deduce that $1/1.3-1/3.5+1/5.7-1/7.9+.....=[\pi-2] / [4]$.

(b) Find the Fourier expansion of $x\sin x$ as Fourier cosine series in the interval $[0,\pi]$.

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